Type MS · Read or estimate a value from a graph, chart, table or image

Answer 1 of 11 · Question 6 part b.i · 1 mark

2025 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

6	(a)	(i)	any two from: • alcohol absorbed/passes into blood; • rapidly/immediately/straight away/steep increase; • by stomach;	ignore intestines	2
		(ii)	• (alcohol carried to) liver; • broken down/detoxified; • level (in blood) falls;		3
	(b)	(i)	• any time after (including) 2.4 hours/2 hours 24 mins;	allow a time of between 1-20 minutes	1
		(ii)	• alcohol is a depressant; • slows reaction/increases reaction time; • drivers can't respond as quickly/less alert; • leads to accidents;		4
	(c)		• collecting duct less permeable (in absence of ADH); • less water reabsorbed; • into blood; • volume of urine (therefore) increased/more urine produced; • and is more dilute/less concentrated;	accept ref fewer aquaporins/water channels	5
					15 marks

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 1 of 11 · Question 6 part b.i · 1 mark

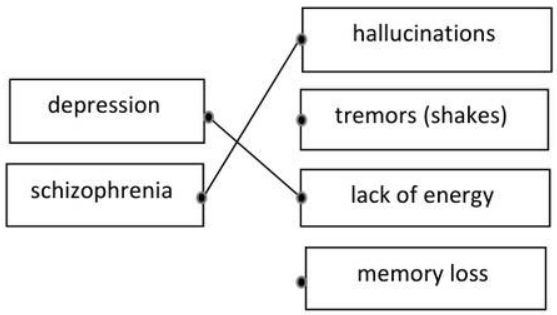
2025 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 2 of 11 · Question 2 part a.i · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

2 (a) (i)	2002;		1
(ii)	2005 (1.4)		1
(iii)	2.75:2.6; 1.1/1.06 :1;		2
(iv)	low income/low self esteem/(domestic) violence/low social status/substance abuse;		1
(b)			2
(c)	**A** alcohol and heroin; B antibiotics have no effect on mental health C paracetamol has no effect on mental health D antibiotics have no effect on mental health		1
(d)	Any two from the following brain/skin/sweat glands/muscles/blood vessels/arterioles;;		2
			Total 10

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 2 of 11 · Question 2 part a.i · 1 mark

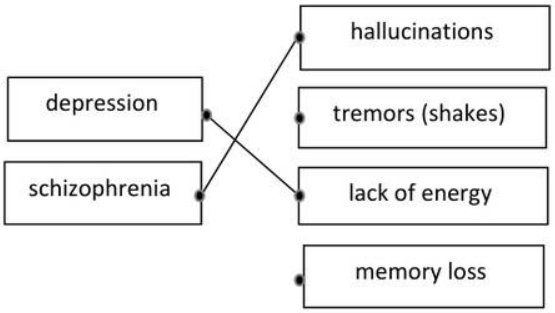
2022 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 3 of 11 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

2 (a) (i)	2002;		1
(ii)	2005 (1.4)		1
(iii)	2.75:2.6; 1.1/1.06 :1;		2
(iv)	low income/low self esteem/(domestic) violence/low social status/substance abuse;		1
(b)			2
(c)	**A** alcohol and heroin; B antibiotics have no effect on mental health C paracetamol has no effect on mental health D antibiotics have no effect on mental health		1
(d)	Any two from the following brain/skin/sweat glands/muscles/blood vessels/arterioles;;		2
			Total 10

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 3 of 11 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 4 of 11 · Question 8 part c.i · 1 mark

2023 JAN · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

Number					
8	(a)	(i)	triggers ovulation/causes release of the egg;		1
		(ii)	Three from: • FSH stimulates growth/production of follicles in the ovary; • more eggs mature/produced; • increases the chances of more than one egg being released; • each egg is fertilised by a sperm;		3
	(b)	(i)	Three from: • placenta supplies glucose/oxygen from the mother; • twins have to share the supply of glucose/receive less glucose/oxygen; • less aerobic respiration; • less energy is released in the foetus and less growth occurs;	Allow nutrients for mp's 1 and 2	3
		(ii)	Two from • same egg and sperm used; • zygote splits/divides • forming 2 embryos;		2
	(c)	(i)	75th		1
		(ii)	• baby is weighed regularly; • growth should follow a percentile/graph/compare with (growth) chart/graph/compare mass to others of the same age; • detect if the baby gains or loses too much mass;		3

Total 13 marks

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 4 of 11 · Question 8 part c.i · 1 mark

2023 JAN · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

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Type MS · Read or estimate a value from a graph, chart, table or image

Answer 5 of 11 · Question 4 part c.i · 1 mark

2023 JUN · Paper 01 · extract 1 of 2 · find part c.i in this mark scheme extract

Number				
4	(a)	any two from • hearing/sound; • touch; • taste; • smell;		2
	(b) (i)	any two from • as thickness decreases, focal length increases; • interval between focal lengths increases as thickness decreases; • (constant when two variables multiplied hence) inversely proportional;		2
	(ii)	diameter decreases;		1
	(iii)	any four from • rays from near object diverging; • ciliary muscle contracts; • suspensory ligaments slacken/relax; • so lens becomes thicker/more convex; • more powerful lens/light refracted more/increased refraction; • (allows it to focus because) focal length becomes shorter/decrease;		4
	(c) (i)	between 575 and 580 nm;		1
	(ii)	• 400 - 530; • 130 nm;	ECF	2
	(iii)	• greater range of wavelengths can be seen/absorb light of different wavelengths; • gives better colour perception/vision/differentiation/can see different colours;		2

Total 14

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 5 of 11 · Question 4 part c.i · 1 mark

2023 JUN · Paper 01 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 6 of 11 · Question 4 part a.ii · 2 marks

2023 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

<table>
<tr>
<td>4 (a) (i)</td>
<td>
<table>
<tr><th>Activity</th><th>Pulse rate
in beats per minute</th></tr>
<tr><td>walking</td><td></td></tr>
<tr><td>jogging</td><td></td></tr>
<tr><td>running</td><td></td></tr>
</table>
</td>
<td>1 mark for valid table
1 mark for labels;
1 mark for units;</td>
<td>3</td>
</tr>
<tr>
<td>(ii)</td>
<td>
<table>
<tr><th>Activity</th><th>Pulse rate
in beats per minute</th></tr>
<tr><td>walking</td><td>94</td></tr>
<tr><td>jogging</td><td>116</td></tr>
<tr><td>running</td><td>133.5</td></tr>
</table>
</td>
<td>2 marks for all readings correct.
1 mark for 1 or 2 readings correct.

tolerance +/- ½ square</td>
<td>2</td>
</tr>
<tr>
<td>(iii)</td>
<td>Any four from five

• take resting pulse;
• method for measuring pulse rate e.g. finger on wrist/neck/electronic device;
• carry out an activity/exercise (for a set period of time/given time);
• measure pulse rate after activity;
• repeat (for each activity);</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>(b)</td>
<td>Any four from

• faster/more/greater muscle contraction during running;
• more energy/ATP required;
• faster rate of respiration;
• more oxygen/ glucose required/more CO_2 removed;
• increase blood flow/faster blood flow/more blood (to muscles);</td>
<td>ORA for all marking points</td>
<td>4</td>
</tr>
</table>

Total for Question 4 = 13 marks

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 7 of 11 · Question 3 part a.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number					
3	(a)	(i)	• time in minutes; • 0-10, 11-20, 21-30, 31-40,41-50,51-60;		2
		(ii)	• 75 + 72 + 68; plus/minus 2 for each one • 215 cm^3;	ECF	2
		(iii)	• rapid reaction/a lot/large amount of gas produced, at start; • rate slows/less gas (for next 10 minutes); • drops a lot/little gas produced (after 40 minutes);		3
		(iv)	any three from • at start plenty of substrate/many E-S complexes formed; • substrate used up/less substrate available; • reaction slows/less gas/product, produced/less enzyme action; • most substrate used at end so little gas produced;		3
	(b)		• less gas produced/reaction lower/less; • because active sites blocked;		2

Total 12

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 7 of 11 · Question 3 part a.ii · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 8 of 11 · Question 2 part d.i · 1 mark

2020 JAN · Paper 01 · find part d.i in this mark scheme extract

number				
2 (a)	(i)	**The only correct answer is A** *B is incorrect because it is the cervix* *C is incorrect because it is the oviduct* *D is incorrect because it is the uterus wall*		1
	(ii)	**The only correct answer is D** *A is incorrect because it is the vagina* *B is incorrect because it is the cervix* *C is incorrect because it is the oviduct*		1
	(iii)	• oestrogen (1) • lining (1) • ovulation (1)		3
(b)		needed for the growth of bones (1)		1
(c)		ref antibodies (passed to baby)/contains the correct balance of nutrients/reduced risk of diabetes/allergies/obesity later in life/reference to bonding between mother and baby (1)		1
(d)	(i)	more breastfed babies in 2010 compared with 1940 (1)	Allow reverse argument A 78% vs 42% difference of 36%	1
	(ii)	• life expectancy shows a constant increase (1) • despite number of babies breastfed increasing and decreasing (1)		2

Total for Question 2 = 10 marks

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 9 of 11 · Question 4 part a.i · 1 mark

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

4(a)(i)	¼/25%/1:4/0.25;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	½/50%/1:2/0.5/2 4;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(b)	head decreases (in proportion); legs increase (in proportion);		**(2)**

Question Number	Answer	Additionalguidance	Mark
4(c)	• allowance for babies of different height; • allows comparison to norm (in population); • determine changes in proportion; • over time / indicates/shows which parts of body are developing/growing;		**Max (2)**

Total 6 Marks

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 9 of 11 · Question 4 part a.i · 1 mark

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 10 of 11 · Question 4 part a.ii · 1 mark

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

4(a)(i)	¼/25%/1:4/0.25;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	½/50%/1:2/0.5/2 4;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(b)	head decreases (in proportion); legs increase (in proportion);		**(2)**

Question Number	Answer	Additionalguidance	Mark
4(c)	• allowance for babies of different height; • allows comparison to norm (in population); • determine changes in proportion; • over time / indicates/shows which parts of body are developing/growing;		**Max (2)**

Total 6 Marks

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 10 of 11 · Question 4 part a.ii · 1 mark

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 11 of 11 · Question 3 part b · 1 mark

2020 JAN · Paper 02 · find part b in this mark scheme extract

Question number					
3	(a)	(i)	**The only correct answer is C** lens *A is incorrect because this is where the image is formed* ***B** is incorrect because that is what controls the size of the pupil* ***D** is incorrect because the pupil is the hole in the middle of the iris*		1
		(ii)	• retina(1) • pupil (1) • choroid(1)		3
	(b)		Value between 190 and 230 (μm)		1
	(c)	(i)	• transmit (nerve/electrical) impulses(1) • to the brain/CNS/relay neurones(1)		2
		(ii)	• less/no impulses able to reach the brain(1) • vision blurred/less clear/less focussed/image not formed/seen/reference to blindness(1)	Allow signals	2

Total for Question 3 = 9 marks